

HAT-P-36b

R-0773

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_250531 · created 2026-07-28T05:25:53+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460826.774 +/- 0.0064 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.138 +/- 0.044
Transit depth (Rp/Rs)^2	1.91 +/- 1.23 [%]
Orbital Inclination (inc)	89.98 +/- 2.83
Airmass coefficient 1 (a1)	0.947 +/- 0.059
Airmass coefficient 2 (a2)	0.06 +/- 0.13
Scatter in the residuals of the lightcurve fit is	6.26 %
Best Comparison Star	#1 - [419, 133]
Optimal Aperture	2.95
Optimal Annulus	16.01
Transit Duration (day)	0.0619 +/- 0.0069

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.138 $\pm$ 0.044 vs 0.126 (deviation 0.27 σ)
Duration fit vs published	0.0619 d vs 0.0925 d (deviation 4.43 σ)
Mid-time O-C	-15.5 min
Depth SNR	3.1
Residual scatter	6.26 %

Light curve

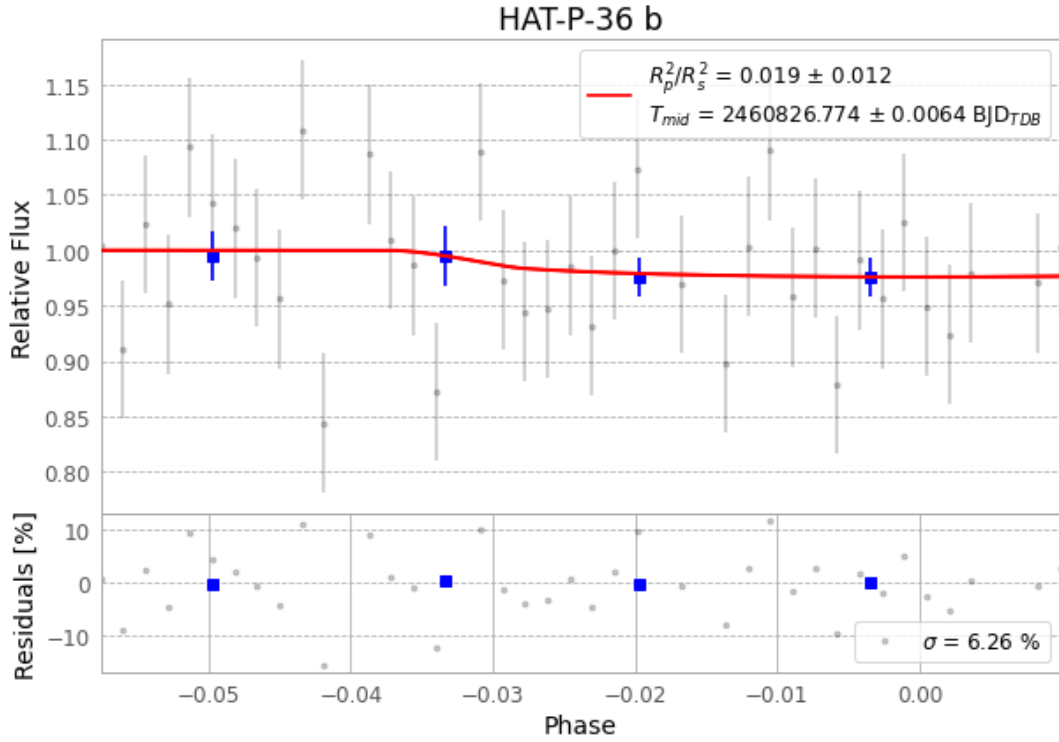


Figure 1 — FinalLightCurve_HAT-P-36 b_2025-05-30.png (SHA-256 8508de654a1a091e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [313, 263], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 01f5f81bc3e9ef43...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

47 FITS frames (31 MB), HATP-36250531044215.FITS → HATP-36250531070015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-250531.log (7b96686364f0...), FinalLightCurve_HAT-P-36 b_2025-05-30.png (8508de654a1a...), FinalLightCurve_HAT-P-36 b_2025-05-30.csv (3c1973a968a0...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 05:25Z.